

# Thanh Dang

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## EDUCATION

**California State Polytechnic University, Pomona**

Bachelor of Science, Computer Science

**August 2022 – May 2026**

**GPA: 3.73**

- **Honors:** Presidential and Dean's List: 2022-2026

## SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- **Frameworks:** React, React Native, Next.js, NestJS, Flask, Ray, PyTorch
- **Tools & Technologies:** Firebase, AWS, PostgreSQL, MySQL, Docker, Kafka, Redis, Node.js, Git

## EXPERIENCE

AI/ML Researcher & Full Stack Developer, **CPP Enterprises**, Pomona, CA

**May 2025 – Present**

- Developed GPU-enabled video processing service using Kafka for event streaming and Ray to orchestrate multi-step, asynchronous pipelines, delivering real-time updates via Server-Sent Events and Redis pub/sub
- Implemented NestJS REST API serving multiple features including route design, service-layer logic, and PostgreSQL persistence; deployed on Jetstream2 via CI/CD pipeline and Docker Compose orchestration
- Spearheaded development of audio-based task identification using WhisperX transcription with speaker diarization and Google Gemma LLM few-shot prompting, processing 50+ hours of mobility scooter driving footage
- Trained multi-task Transformer model on pose sequences to analyze bilateral symmetry and predict side weakness, achieving 91% symmetry separation and 92% AUC for weak-side classification across clinical sessions

Undergraduate Research Assistant, **Cal Poly Pomona**, Pomona, CA

**October 2024 – May 2025**

- Built Python desktop application for Delta NCSA HPC cluster file management via SFTP, streamlining upload, download, and deletion workflows; presented at Cal Poly Pomona RSCA Conference to 20+ attendees
- Utilized YOLO11n-pose model to extract normalized upper-body keypoints from front-view videos, implementing IoU-based person tracking, feeding into downstream stability classification and sway displacement analysis
- Aligned label, motion, and pose data across 40 recording sessions by filtering non-driving segments through temporal segmentation, reducing overall dataset size by 15% while retaining 100% of relevant driving events

Code Coach, **theCoderSchool**, Cerritos, CA

**January 2024 – February 2025**

- Led coding classes for 30+ elementary students with no prior programming experience, teaching fundamental programming concepts and guiding students to build functional platformers and dialogue-driven games in Scratch
- Coached 5 students one-on-one in front-end web development, advancing from core HTML and CSS layout principles to designing and building interactive web pages with JavaScript, leveraging Replit for live deployment

## PROJECTS

TrailTalk

**September 2025 – December 2025**

- Launched social media web app for park enthusiasts using React and Firebase in Agile team environment, featuring 63 U.S. national parks with dedicated park pages, user reviews, photo sharing, and following features
- Designed scalable class-based NoSQL data model supporting user accounts, messaging, and content management; integrated MapLibre for interactive mapping and geolocation features across 50 states
- Implemented live private messaging system with TypeScript and Cloud Firestore, incorporating notifications and read receipts to enable seamless user-to-user communication with low-latency message delivery

Carbon Closet, **Cal Poly Pomona Hackathon 2025**

**April 2025**

- Built full-stack gamified e-commerce platform that makes buying, selling, and donating secondhand clothing simple and rewarding through token-based rewards; presented to 8 judges and won 3rd place among 60 teams
- Developed responsive interface with Next.js and Cloud Firestore, supporting live listing updates, real-time search filtering, shopping cart management, user profiles, and gacha bundle purchases with dynamic inventory tracking
- Containerized application with Docker to ensure consistent environments across development and production; deployed to AWS EC2, gaining hands-on experience with cloud hosting and production deployment workflows

Recyclable or Not Bot, **UC Riverside Hackathon 2025**

**January 2025**

- Created computer vision web app to detect recyclable vs. non-recyclable items, rendering real-time classification, color-coded score to promote eco-friendly waste disposal; awarded best sustainability track among 50+ teams
- Integrated end-to-end pipeline using YOLOv8 deep learning model for object detection, with Flask API managing OpenCV frame capture and delivering inference results to React front-end alongside recycling tips and resources

## EXTRACURRICULARS

**STEM Advantage Scholar:** Scholarship program for first-generation college students in STEM

**June 2024 – May 2025**

**Software Engineer Association:** Largest software engineering community at CPP

**November 2023 – May 2024**